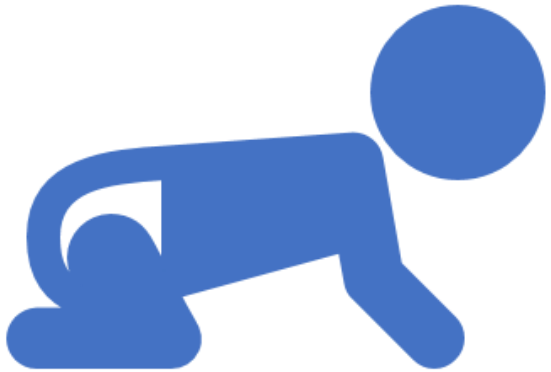
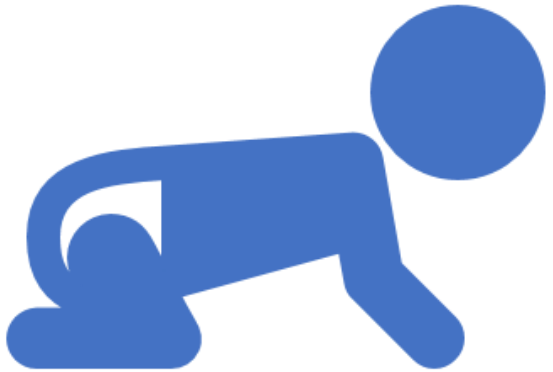


Vaccination of Preterm Infants

Adel Reyad

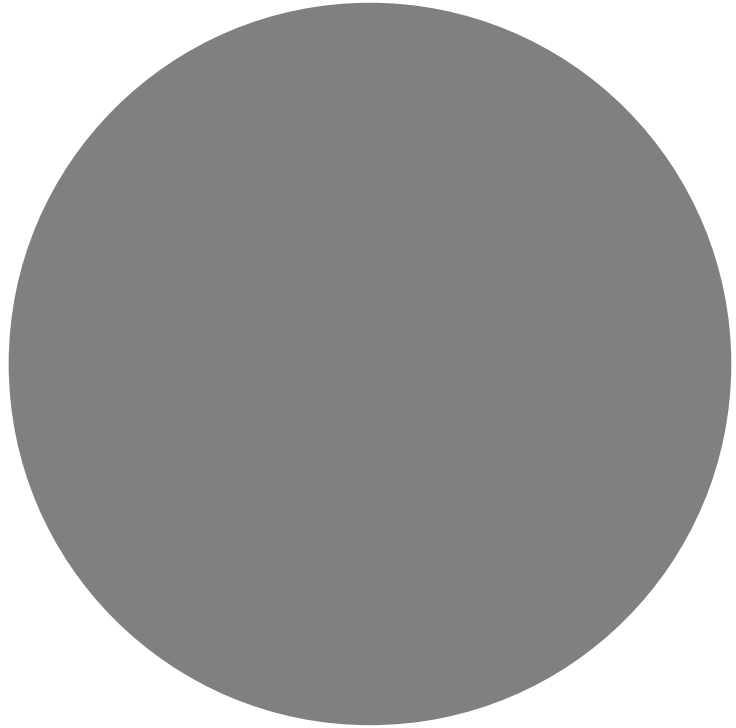


Vaccination of Preterm Infants

Adel Reyad

Overview

- **Preterm** (premature) infants are those born at <37 weeks gestation.
- **Extremely preterm** infants are born at <28 weeks gestation.



Overview

- Prematurity can increase the child's risk of **vaccine-preventable diseases**.[1-3]

Overview

Despite their immunological immaturity,

preterm infants generally

respond well to vaccines.[4-6]

Overview

Provided they are

***medically stable**

***no contraindications to vaccination**[7-9]

Overview

Preterm infants should receive vaccines according to

***recommended schedule**

***chronological age,**

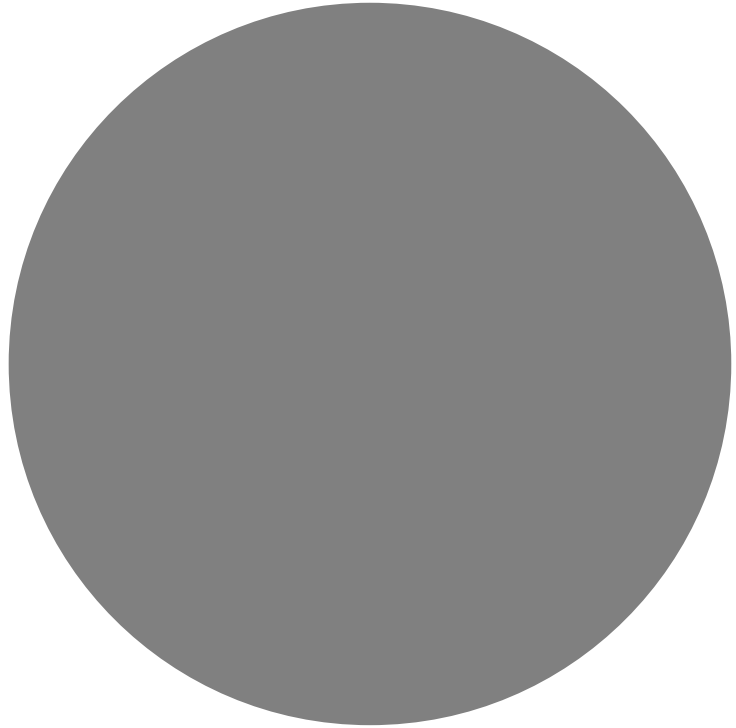
without correction for prematurity.[7-9]

- Before giving vaccines, **CONSIDER**

*Child’s birth weight

*Precise gestational age

*Presence of any chronic medical condition(s)

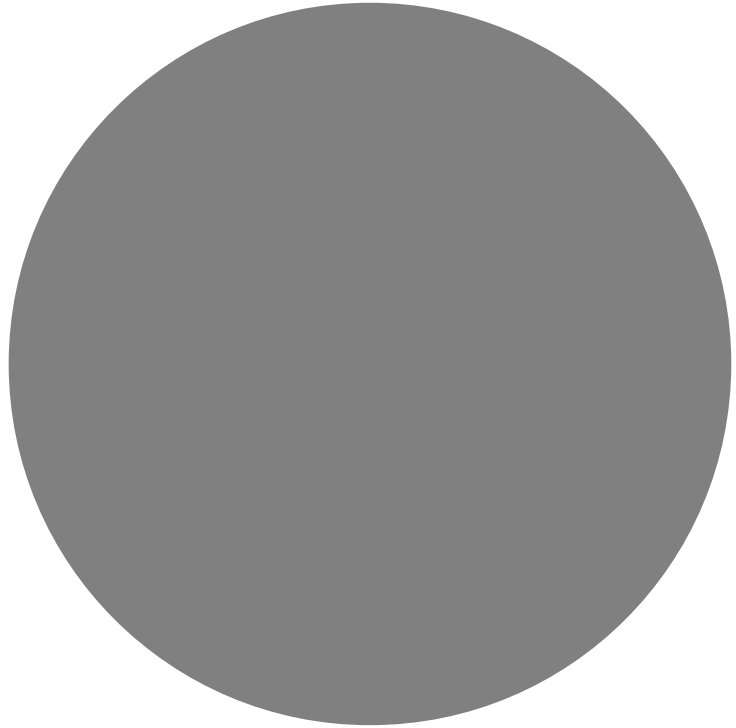


Overview



Apnea and Vaccination

- Vaccination is associated with an increased risk of apnea in preterm infants who receive vaccines in hospital,
- particularly those receiving complex medical care or with a history of apnea.



Managing Apnea after Vaccination of preterm Infants

- Apnoea is generally self-limiting in this setting,

but is an expected adverse event following immunisation. [10-12]

Managing Apnea after Vaccination of preterm Infants

- **<u>Take measures to manage it by</u>**:

monitoring hospitalised preterm infants

for apnea or bradycardia for up to

48 hours after vaccination[12,13]

Managing Apnea after Vaccination of preterm Infants

- In infants with a history of apnea after vaccination, considering giving **future vaccines under medical supervision**[14,15]

Managing Apnea after Vaccination of preterm Infants

- Vaccination is **NOT ASSOCIATED** with an increased risk of **SUDDEN INFANT DEATH SYNDROME**.[16,17]

Managing Apnea after Vaccination of preterm Infants

Pneumococcal Vaccine (PCV)



Pneumococcal Vaccine (PCV)



- **All preterm infants born at <28 weeks gestation**

are recommended to receive

- **4 doses of PCV (conjugate vaccine)**
- **at 2, 4, 6 and 12 months of age.**

Pneumococcal vaccine

- **A single booster dose of 23vPPV**

(23-valent Pneumococcal)

(Polysaccharide Vaccine)

at 4–5 years of age is also recommended.

Pneumococcal vaccine

- **<u>Children who were born at <28 weeks gestation</u>**

No need for PCV doses after 5 years of age if they:

- 1- Do not have a chronic medical condition(s)

- 2- Have received the extra pneumococcal vaccine at 5 years of age

Pneumococcal vaccine

- All children and adults who have:
- **Chronic lung disease** or
- **Chronic medical conditions**

(whether related to preterm birth or not)

Pneumococcal vaccine

should receive extra pneumococcal

vaccine doses up to and beyond

the age of 5 years.

_____

Pneumococcal vaccine

Hepatitis B vaccine

Hepatitis B vaccine

- **Low-birthweight preterm infants**

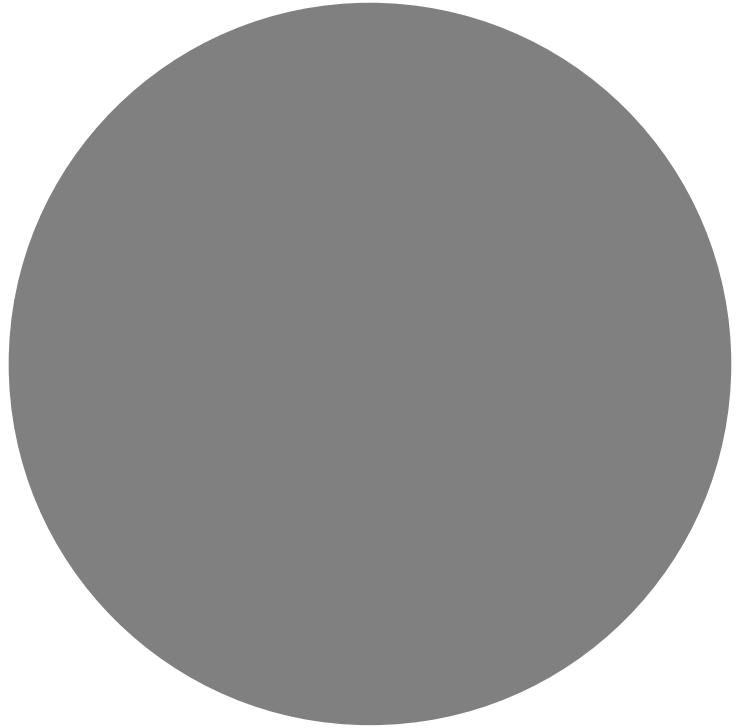
do not respond well to hepatitis B–containing

vaccines as full-term infants do .[13,18,19]

Hepatitis B vaccine

Hepatitis B vaccine

- **<u>Infants (<2000 g) and/or infants <32 weeks</u>** (irrespective of weight) should receive:
- Hepatitis B vaccine at birth
- 3 doses of a hepatitis B–containing vaccine, at 2, 4 and 6 months of age
- A booster dose at 12 months of age



Hepatitis B vaccine

There is no need to measure the hepatitis B surface antibody (HBs) titre after the primary series.

Hepatitis B vaccine

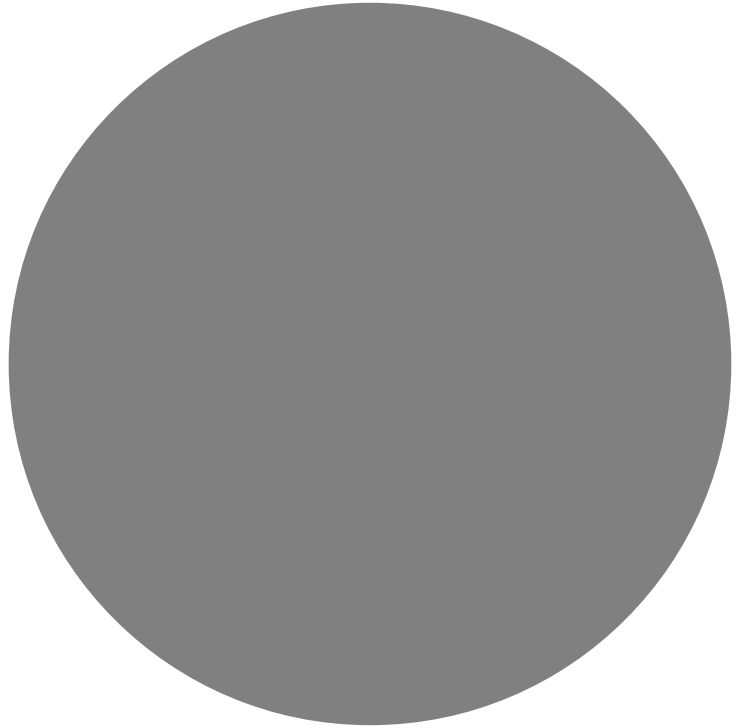
If measuring the antibody titre,

do it at least 1 month after the 6-month dose.

If <10 mIU / mL, give a **booster dose.**

- **Preterm infants of (HBsAg)–positive mothers**

- They should receive hepatitis B vaccine and **hepatitis B immunoglobulin**.

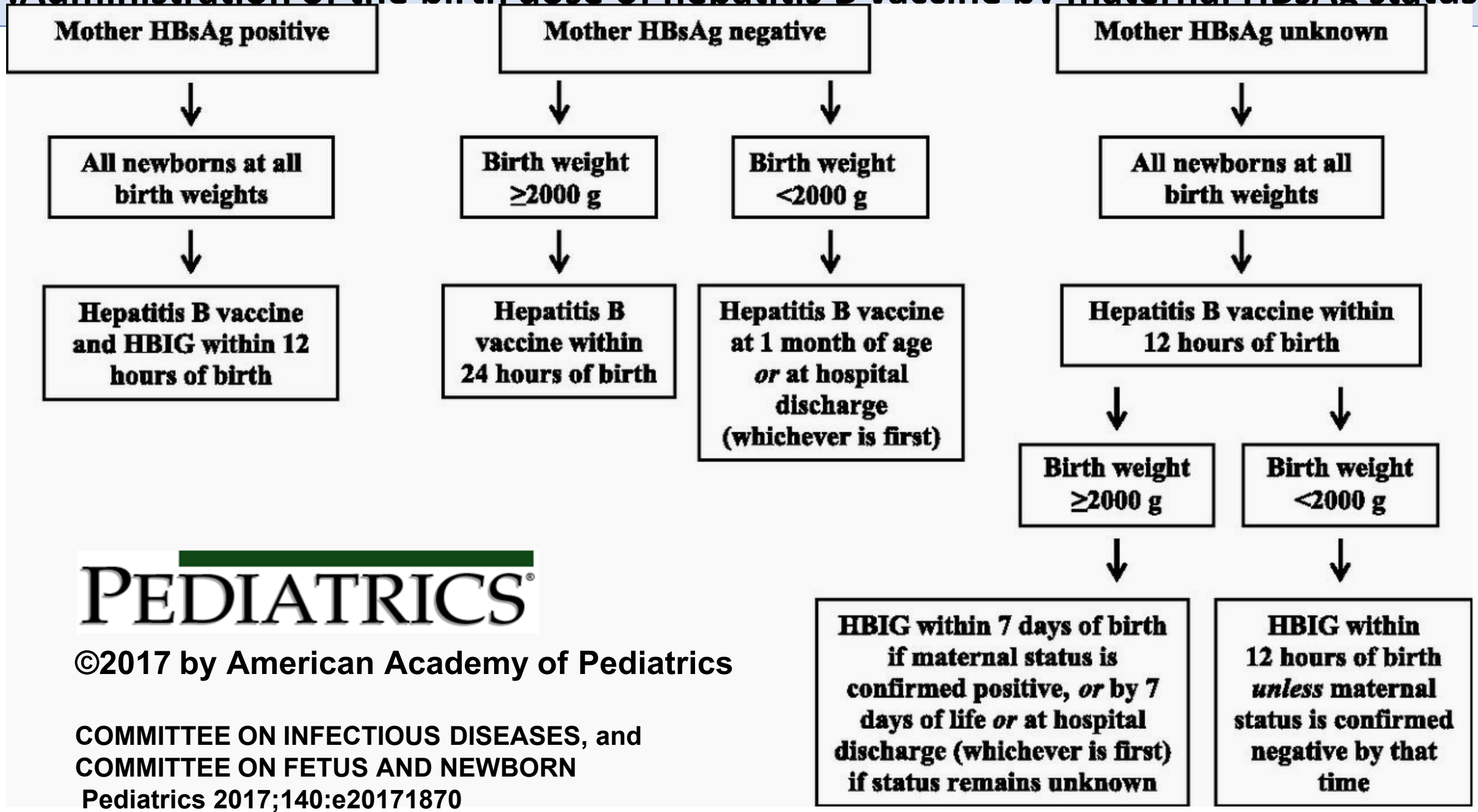


Hepatitis B vaccine

In October 2016, the ACIP rescinded the permissive language and instead stated the following:

- "For all medically stable infants weighing ≥ 2000 grams at birth and born to HBsAg-negative mothers, **the first dose of vaccine should be administered within 24 hours of birth.**
- Only single-antigen hepatitis B vaccine should be used for the birth dose."[13,14]

.Administration of the birth dose of hepatitis B vaccine by maternal HBsAg status



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COMMITTEE ON INFECTIOUS DISEASES, and
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Pediatrics 2017;140:e20171870

After completion of a 3- or 4-dose hepatitis B vaccine series, 98% of healthy term infants achieve protective antibody concentrations.

Protection may be lower in infants with birth weights less than 2000 g.[5]

Vaccine Issues

Vaccine Issues

The optimal time for serologic testing to detect a vaccine response is

1 - 2 months after the final dose of the vaccine series.

Vaccine Issues

Recommendations for postvaccination serologic testing

of infants born to hepatitis B–infected mothers

have been updated recently.[16]

Vaccine Issues

Postvaccination serologic testing by

1- Measuring hepatitis B surface antibody (anti-HBs) to document protection

2- HBsAg to rule out perinatal infection

now is recommended at 9-12 m instead of 9-18 m.

Vaccine Issues

Infants who are HBsAg negative

&

anti-HBs levels <10 mIU/mL

(nonprotective)

REQUIRE ADDITIONAL VACCINE DOSES.

Vaccine Issues



ACIP considered revaccination strategies in February 2017 and recommended

- HBsAg-negative infants with anti-HBs levels <10 mIU/mL **receive a single additional dose** of hepatitis B vaccine and be **retested** for anti-HBs 1 to 2 months after that dose.

Vaccine Issues



- **If the anti-HBs level is still <10 mIU/mL,**

the infant should receive

- **2 additional doses of hepatitis B vaccine**
- followed by **retesting**

1- 2 months later.

Vaccine Issues



- Infants with anti-HBs levels less than 10 mIU/mL after 2, 3-dose series of hepatitis B vaccine are **non-responders,** available data do not suggest benefit from additional vaccinations.

Vaccine Issues



ACIP noted

- **On the basis of clinical circumstance or family preference,**

HBsAg-negative infants with anti-HBs levels <10 mIU/mL

may be vaccinated instead with a

second complete series

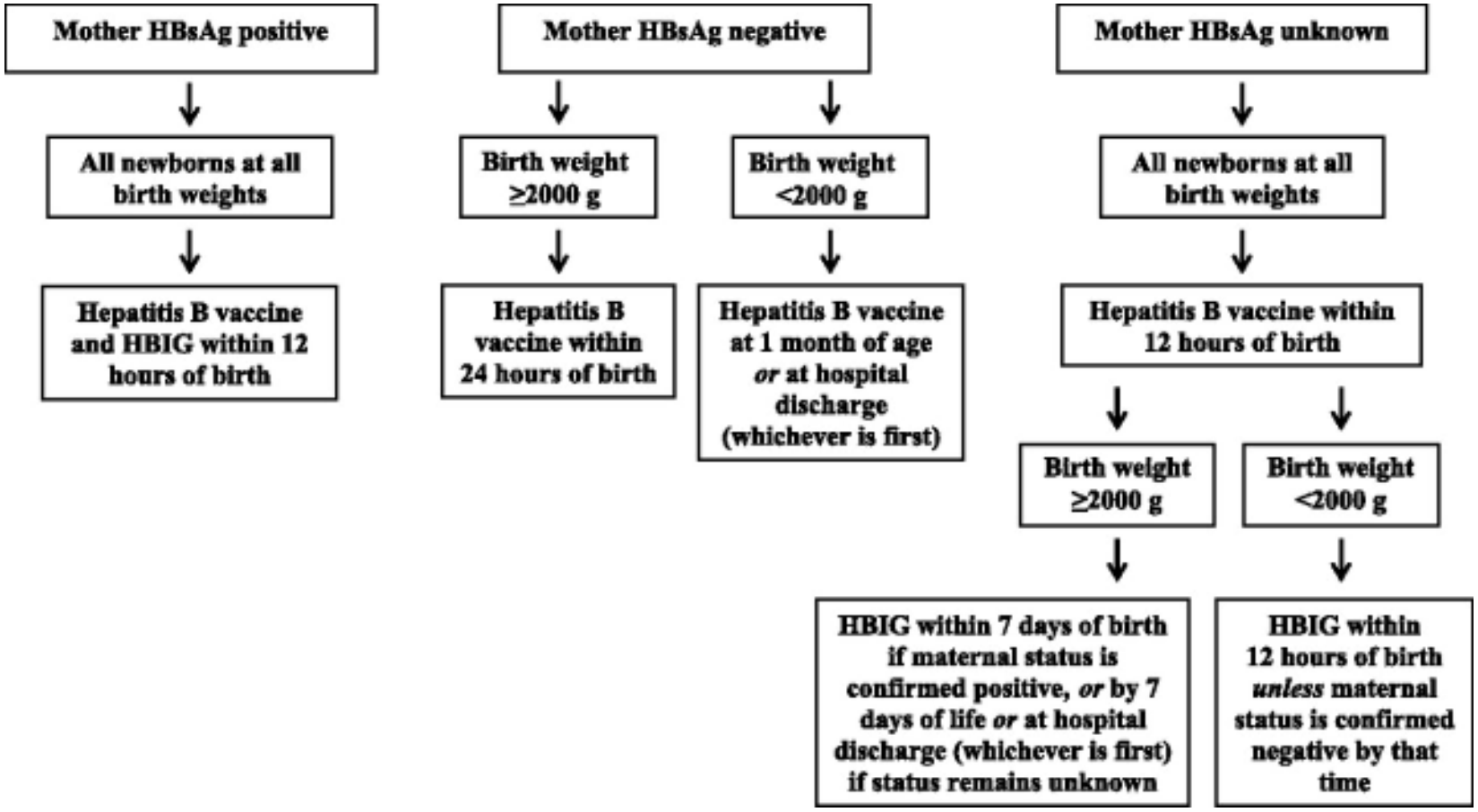
followed by postvaccination retesting

1-2 months after the final dose.

Vaccine Issues

- After completion of a 3- or 4-dose hepatitis B vaccine series, 98% of healthy term infants achieve protective antibody concentrations.
- Protection may be lower in infants with birth weights less than 2000 g.[5]
- The optimal time to perform serologic testing to detect a vaccine response in infants is 1 to 2 months after the final dose of the hepatitis B vaccine series.
- Recommendations for postvaccination serologic testing of infants born to hepatitis B–infected mothers have been updated recently.[16]
- For infants born to HBsAg-positive mothers, postvaccination serologic testing by measuring hepatitis B surface antibody (anti-HBs) to document protection and HBsAg to rule out perinatal infection now is recommended at 9 to 12 months of age instead of 9 to 18 months.
- Infants who are HBsAg negative with anti-HBs levels less than 10 mIU/mL (nonprotective) require additional vaccine doses.
- The ACIP considered revaccination strategies in February 2017 and recommended that HBsAg-negative infants with anti-HBs levels less than 10 mIU/mL receive a single additional dose of hepatitis B vaccine and be retested for anti-HBs 1 to 2 months after that vaccine dose.
- If the anti-HBs level is still less than 10 mIU/mL, the infant should receive 2 additional doses of hepatitis B vaccine followed by retesting 1 to 2 months later.
- Infants with anti-HBs levels less than 10 mIU/mL after 2 3-dose series of hepatitis B vaccine are nonresponders, and available data do not suggest benefit from additional vaccinations.
- The ACIP noted that, on the basis of clinical circumstance or family preference, HBsAg-negative infants with anti-HBs levels less than 10 mIU/mL may be vaccinated instead with a second complete series followed by postvaccination testing 1 to 2 months after the final dose.

Administration of the birth dose of hepatitis B vaccine by maternal HBsAg status.



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Influenza vaccine



Influenza vaccine

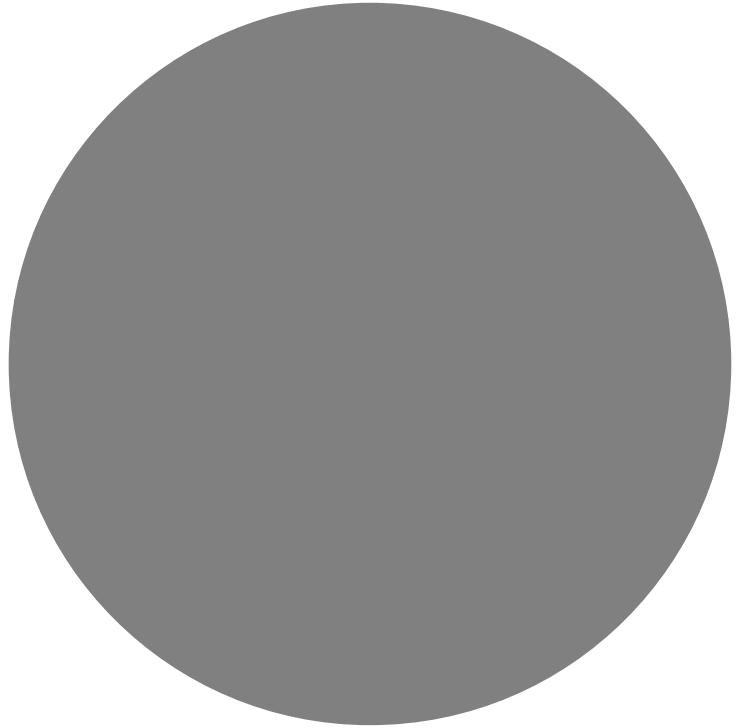
- Preterm infants have an increased risk of complications from influenza.[20]
- Due to underlying respiratory, cardiac and neurologic conditions.

Influenza vaccine

Preterm infants should receive

influenza vaccine every year,

starting at ≥6 months of age.



Influenza vaccine

Preterm infants should receive

influenza vaccine **every year**,

starting **at ≥6 months** of age.

Influenza vaccine

Influenza vaccine

- Infants should receive 2 vaccine doses, at least 4 weeks apart, the 1st time they receive influenza vaccine.
- Then 1 dose every year after that.

Recommended administration of influenza vaccine in

Allergy	Vaccine administration and setting
Uncertain (eg positive skin test but not yet eaten egg)	Vaccinate with full age-appropriate dose in any immunisation setting
Non-anaphylaxis egg allergy	Vaccinate with full age-appropriate dose in any immunisation setting
Anaphylaxis egg allergy	Vaccinate with full age-appropriate dose in a medical facility with staff experienced in recognising and treating anaphylaxis

People with a history of Guillain–Barré syndrome

- People with a history of

Guillain–Barré syndrome (GBS)

whose first episode

was not after influenza vaccination

have an extremely low risk of

recurrence of GBS after vaccination.

Influenza vaccination is recommended

for these people[63-65]

GBS

- Influenza vaccination is generally not recommended for people with a history of GBS whose first episode occurred within 6 weeks of receiving an influenza vaccine.

GBS

recurrence of GBS in people

where the first episode occurred

within

6 weeks of influenza vaccination

(i.e. the first episode was possibly

triggered by the vaccine).

GBS

- In these people, discuss

1- The potential for recurrence if vaccinated,

2- The potential for exacerbation following influenza **infection**,

3- Other protective strategies (e.g. vaccination of household members).

GBS

- Vaccination can be considered in special circumstances,

such as when an alternative cause for GBS,

such as *Campylobacter jejuni* **infection**, was found

or the risk of influenza disease is considered high.

Rotavirus Vaccine



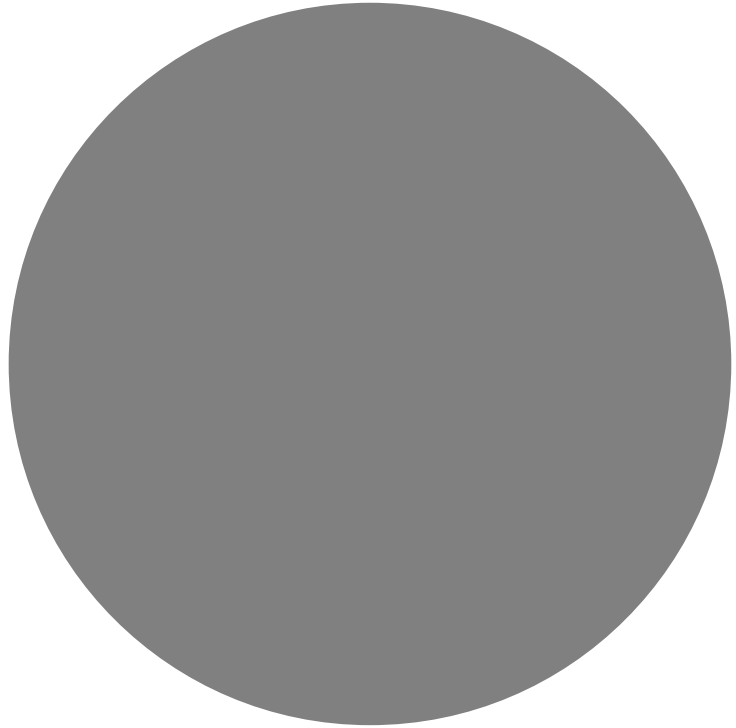
Rotavirus Vaccine



- Preterm infants can receive rotavirus vaccine at their chronological age without correction for prematurity.
- This includes hospitalised infants who are medically stable.

Rotavirus Vaccine

- Strict upper age limits for vaccination apply, depending on which rotavirus vaccine the infant receives.



Rotavirus Vaccine

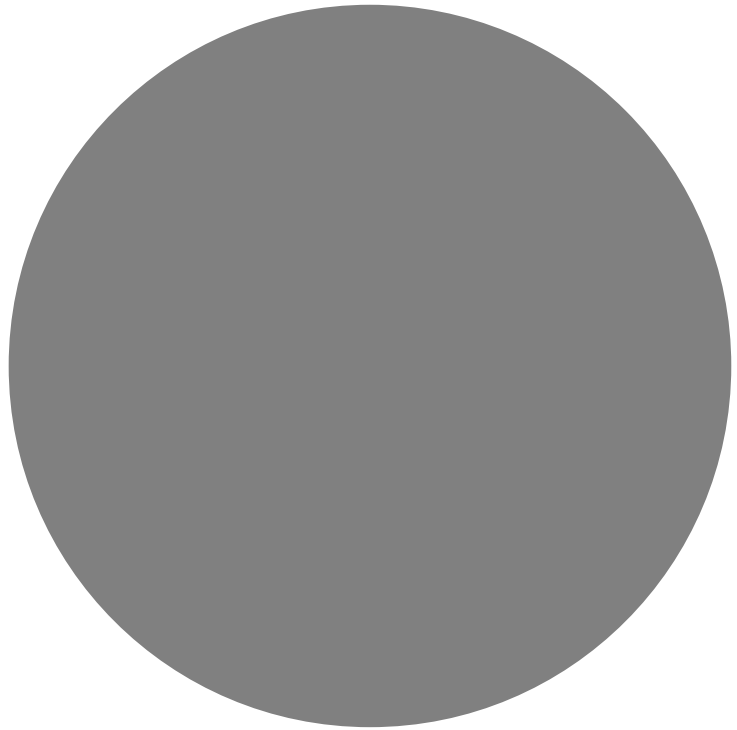
Hib vaccine

Hib vaccine

- Preterm infants do not need a change to the usual schedule for *Haemophilus influenzae* type b (Hib) vaccines conjugated with tetanus (PRP-T), including Infanrix hexa.

Hib vaccine

- Preterm infants produce good antibody responses to all the antigens in Infanrix hexa following administration at 2, 4 and 6 months of age.

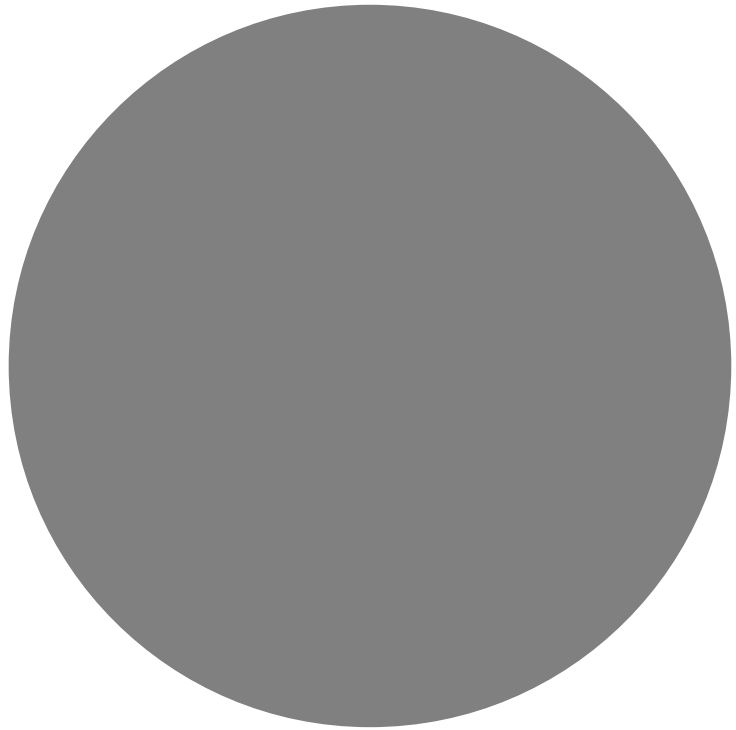


Hib vaccine

- The responses to hepatitis B and Hib

  are not quite as high as in full-term

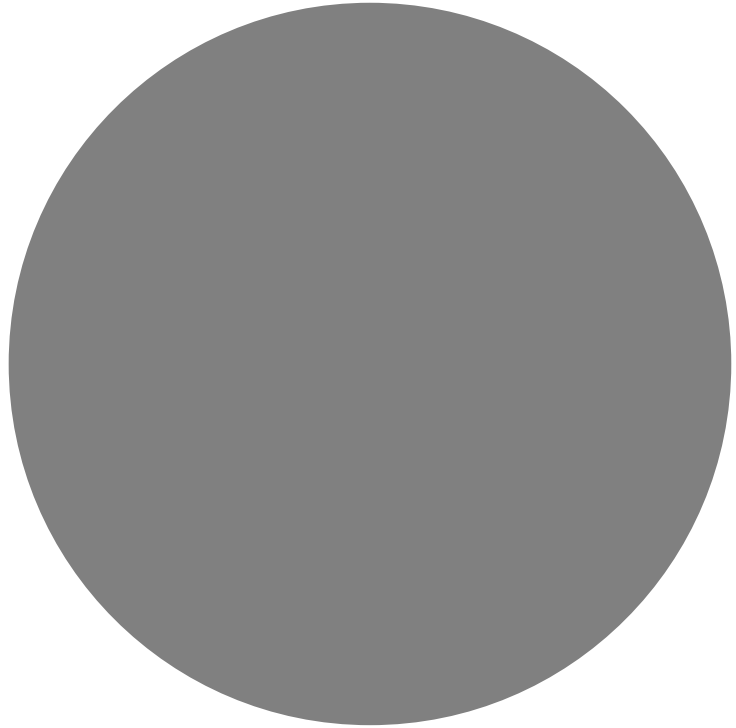
  infants.[1]



Hib vaccine

Hib vaccine

- If an extremely preterm and/or low-birthweight baby (<28 weeks gestation or <1500 g birth weight) received Hib vaccine conjugated to the outer membrane protein of *Neisseria meningitidis* (PRP-OMP) for their primary doses, they should receive an extra dose at 6 months of age.



- That is,

they should receive doses at 2, 4, 6 and 18 months of age.

Hib vaccine

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Thank you